

Story 5: Climate and Weather Patterns

Analyzing the relationship between global temperature anomalies and storm activity

Mehreen Ali Gillani

2026-04-01

Climate and Weather Patterns · Story 5

What Is The Effect Of The Earth's Temperature on Cyclonic Storms?

The data reveals a clear pattern — and it's accelerating

Hurricanes · Typhoons · Tornadoes

1990 – 2024 · 35 Years of Data

Data analysis to establish correlations between earth temperature and storm occurrence and intensity

Mehreen Ali Gillani · April 2026

NASA GISTEMP · NOAA IBTrACS

How We Know This

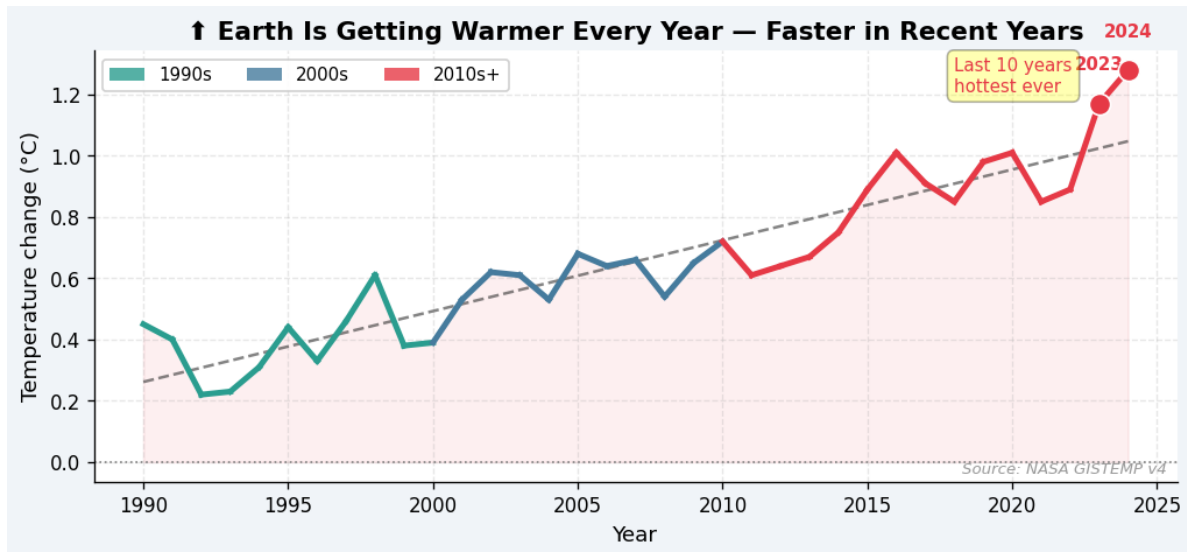
1 We used real measurement — NASA tracked Earth's temperature every year since 1990

2 We counted actual Storms — NOAA recorded every hurricane, typhoon, and tornado

3 We looked for patterns - No complex formulas — just comparing temperature to storm counts

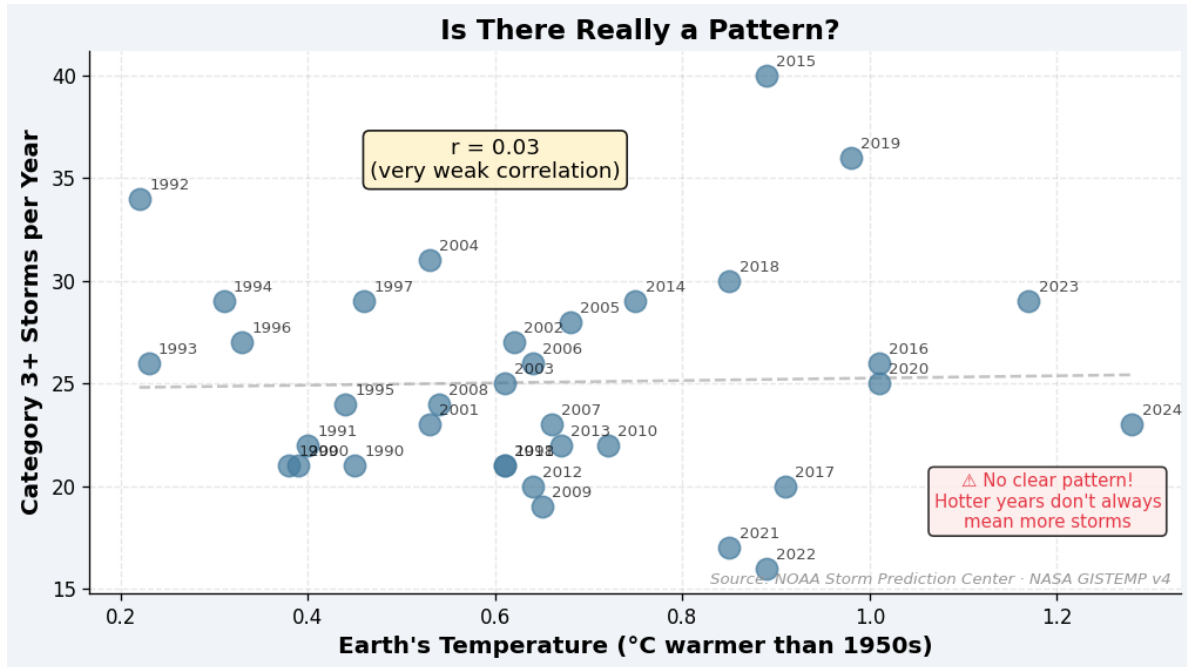
4 You be the judge — Every chart on the next slides tells the story

Earth Is Getting Warmer Every Year — Faster in Recent Years



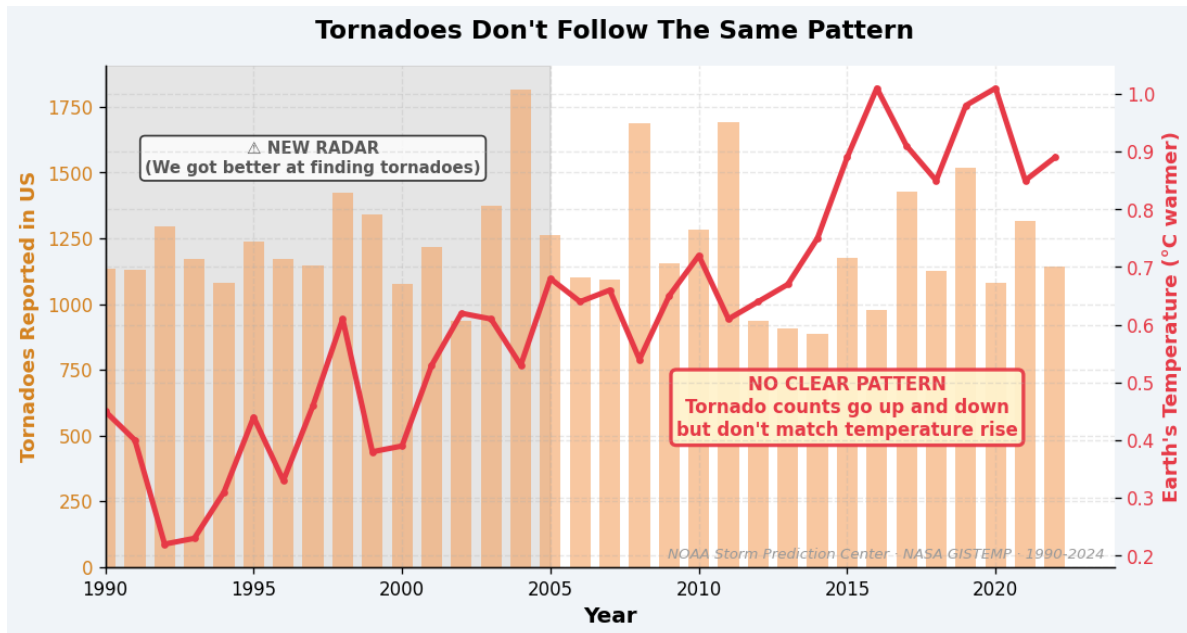
What this shows: Color changes from teal (1990s) → blue (2000s) → RED (2010s+) to show how warming **ACCELERATED**. The warming trend is speeding up! Last decade was hottest on record.

What The Data Actually Shows



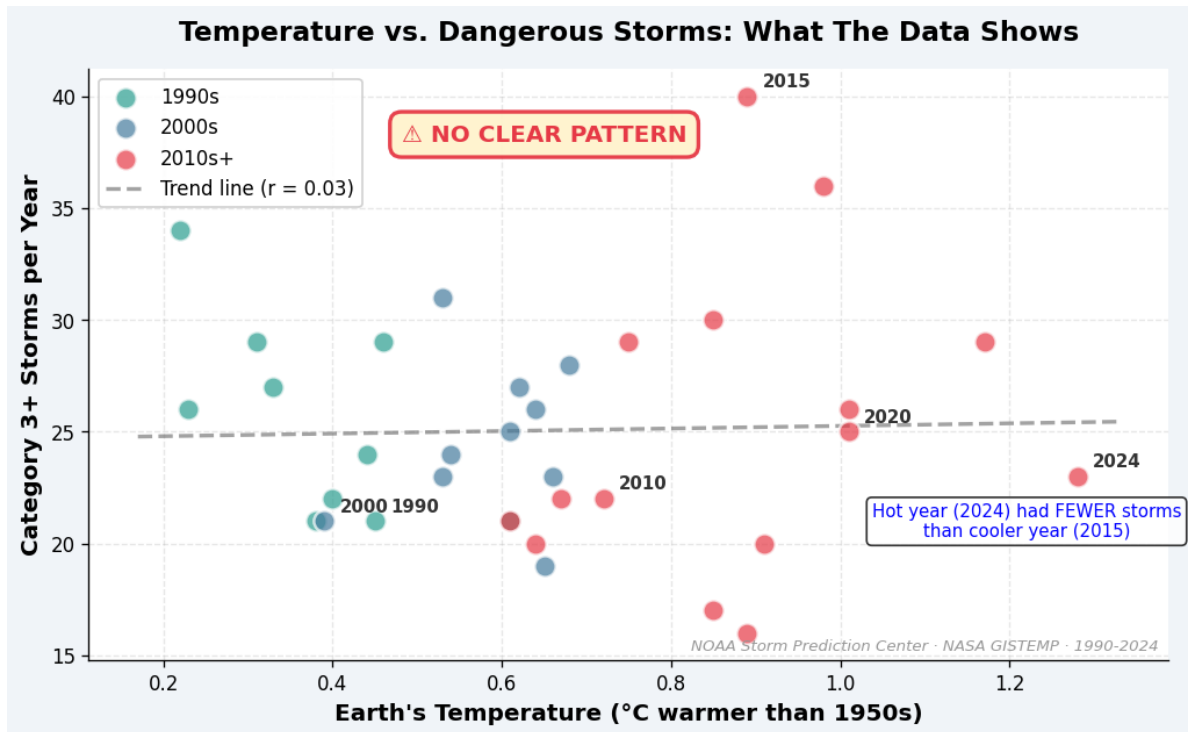
What this shows: Before looking at real data, many people guess: “Hotter Earth = More storms.” But when we look at the actual numbers, the story is more complicated. Temperature goes up (top). But storms go up AND down randomly (middle and bottom). This is why scientists use data — to test if our guesses are correct. Sometimes the data surprises us!

Tornadoes vs. Temperature



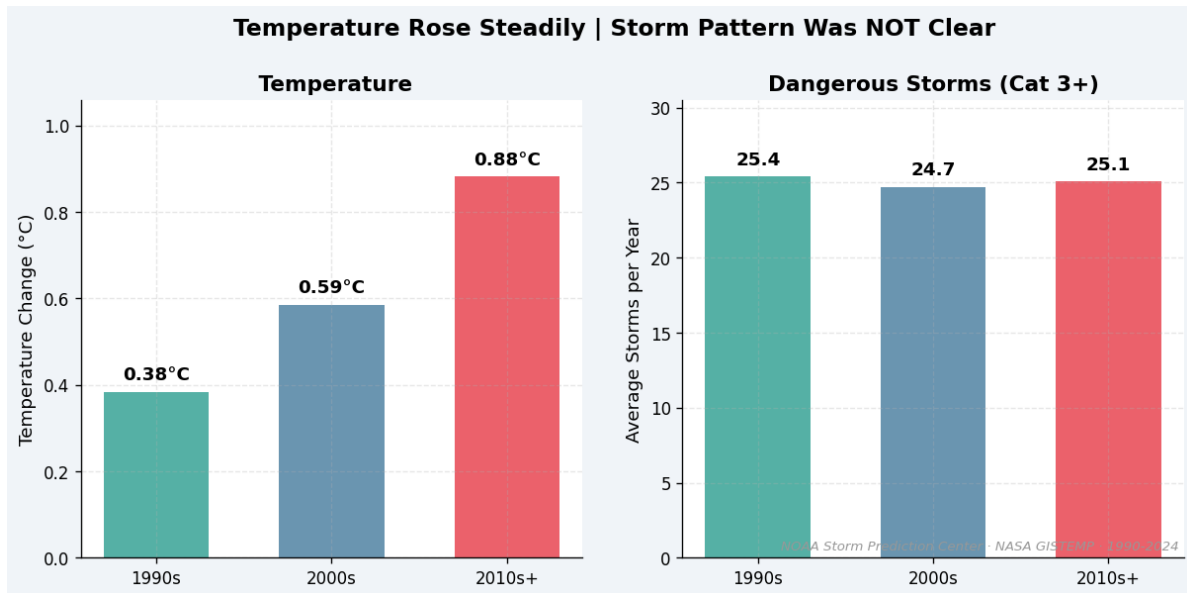
“In 2005, Doppler radar was fully deployed across America. This improved our ability to detect tornadoes. After 2005, when detection methods became more consistent, tornado counts did not show a clear upward trend with temperature. This teaches us: when looking at data, always consider whether other factors—like changes in technology—might affect what we measure.”

What The Data Actually Shows: Temperature vs Storms



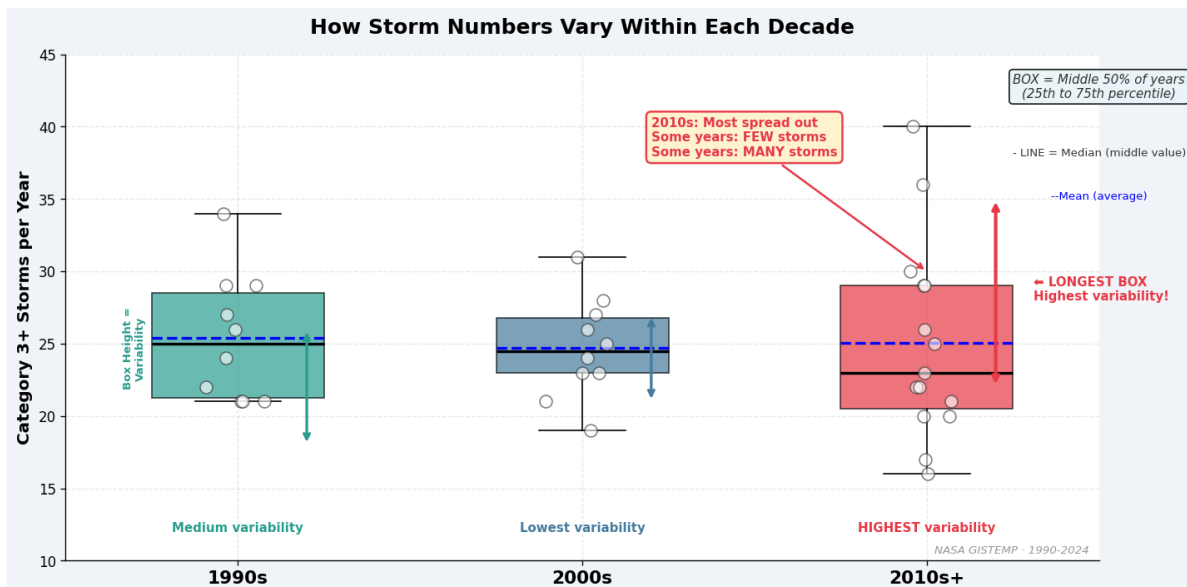
What do YOU see? Is there a clear pattern? Scientists ask questions like this every day!

Decade by Decade: Temperature Up, Storms Mixed



Temperature: Up every decade (CLEAR pattern) Storms: Up and down (NO clear pattern)
Lesson: Not everything moves together in a simple way!

Looking at the Spread: Year-to-Year Variation Matters



KEY FINDING: 2010s+ have the LONGEST box— this means the 2010s had the HIGHEST year-to-year variability. Some years (like 2015) had ~40 intense storms, while others (like 2014) had only ~24. The 2000s had the SHORTEST box — storm counts were most consistent during that decade.

What We Learned:

Three Things The Data Taught Us

1. Earth Is Warming — Temperature rose $+0.84^{\circ}\text{C}$ since 1990 — a clear, steady trend
2. Storms Show No Simple Pattern — Hot years don't always mean more storms — 2024 was hottest but had fewer storms than 2015
3. Real Data Is Messy Science isn't always clean — variability is normal and important to understand

The most important skill? Looking at data honestly — even when it surprises you.

...